



Operating Instructions

Diesel engine
20V4000Gx3x
6 exhaust turbochargers

MS150093/08E

Engine model	Frequency	Application group *
20V4000G23	50 Hz	3B, 3D, 3E, 3F, 3G
20V4000G43	60 Hz	3B, 3D, 3F, 3G
20V4000G63	50 Hz	3A, 3B, 3D, 3E, 3F, 3G
20V4000G63L	50 Hz	3A, 3B, 3D, 3E, 3F, 3G
20V4000G83	60 Hz	3A, 3B, 3D, 3F, 3G
20V4000G83L	60 Hz	3A, 3B, 3D, 3F, 3G
* 3A Heavy duty, unrestricted, ICXN 3B Continuous service, variable load, ICXN 3D Emergency standby power, IFN 3E Emergency service, ICXN 3F Heavy duty for DCP, unrestricted, ICXN 3G Heavy duty, intermittent, ICXN		

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to §15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

© Copyright Rolls-Royce Solutions

This publication is protected by copyright and may not be used in any way, whether in whole or in part, without the prior written consent of Rolls-Royce Solutions. This particularly applies to its reproduction, distribution, editing, translation, microfilming and storage and/or processing in electronic systems including databases and online services.

All information in this publication was the latest information available at the time of going to print. Rolls-Royce Solutions reserves the right to change, delete or supplement the information provided as and when required.

California Proposition 65

This warning applies only to the State of California, USA, in compliance with Title 27 California Code of Regulations, Article 6, Clear and Reasonable Warnings.

 **WARNING:** Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to www.P65warnings.ca.gov/diesel.

(→ www.P65warnings.ca.gov/diesel)

Table of Contents

1 Preface		T2 (40CFR60) and EPA Nonroad T2 Comp (40CFR89)	
1.1 Preface	8	5.8 Product data 20V4000Gx3x, emergency standby power 3D, NEA Singapore for ORDE and China Nonroad Mobile Machinery Stage III	62
2 Safety		5.9 Product data 20V4000Gx3x, emergency service 3E, fuel optimized	66
2.1 Important provisions for all products	9	5.10 Product data 20V4000Gx3x, emergency service 3E, optimized exhaust gas emission (TA-Luft)	69
2.2 Intended use of gendrive engines	10	5.11 Product data 20V4000Gx3x, emergency service 3E, NEA Singapore for ORDE and China Nonroad Mobile Machinery Stage III	72
2.3 Personnel and organizational requirements	11	5.12 Product data 20V4000Gx3x, heavy duty for DCP 3F, fuel optimized	76
2.4 Safety regulations for commissioning and operation	12	5.13 Product data 20V4000Gx3x, heavy duty for DCP 3F, optimized exhaust gas emissions (TA-Luft)	80
2.5 Safety regulations for assembly, maintenance, and repair work	14	5.14 Product data 20V4000Gx3x, heavy duty for DCP 3F, EPA Stationary EMERG T2 (40CFR60) and EPA Nonroad T2 Comp (40CFR89)	83
2.6 Fire and environmental protection, indirect materials	18	5.15 Product data 20V4000Gx3x, heavy duty for DCP 3F, NEA Singapore for ORDE and China Nonroad Mobile Machinery Stage III	87
2.7 Standards for warning messages in the text and highlighted information	21	5.16 Product data 20V4000Gx3x, heavy duty 3G, fuel optimized	91
3 Transport		5.17 Product data 20V4000Gx3x, heavy duty 3G, optimized exhaust gas emissions (TA-Luft)	95
3.1 Transport	22	5.18 Product data 20V4000Gx3x, heavy duty 3G, EPA Nonroad T2 Comp (40CFR89)	98
3.2 Lifting specifications	23	5.19 Product data 20V4000Gx3x, heavy duty 3G, NEA Singapore for ORDE and China Nonroad Mobile Machinery Stage III	101
3.3 Engine mounting transportation lock	24	5.20 Final compression pressure	104
3.4 Transport locking device for crankshaft	26	5.21 Firing order	105
4 General Information		5.22 Engine - Main dimensions	106
4.1 Engine side and cylinder designations	29	6 Operation	
4.2 Engine layout	30	6.1 Preparations for putting into operation after extended out-of-service periods (>3 months)	107
4.3 Sensors and actuators - Overview	31	6.2 Preparations for putting into operation after scheduled out-of-service-period	108
5 Technical Data		6.3 Starting the engine in manual mode (test run)	109
5.1 Product data 20V4000Gx3x, heavy duty 3A, fuel optimized	34	6.4 Re-starting the engine following an automatic safety shutdown	110
5.2 Product data 20V4000Gx3x, continuous service 3B, fuel optimized	38	6.5 Safety system - Override	111
5.3 Product data 20V4000Gx3x, continuous service 3B, optimized exhaust gas emissions (TA-Luft)	43		
5.4 Product data 20V4000Gx3x, continuous service 3B, EPA stationary EMERG T2 (40CFR60) and EPA Nonroad T2 Comp (40CFR89)	46		
5.5 Product data 20V4000Gx3x, continuous service 3B, NEA Singapore for ORDE and China Nonroad Mobile Machinery Stage III	50		
5.6 Product data 20V4000Gx3x, emergency standby power 3D, fuel optimized	54		
5.7 Product data 20V4000Gx3x, emergency standby power 3D, EPA Stationary EMERG	58		

6.6	Emergency engine start (Override mode)	112	9.4.1	Valve protrusion - Measurement	196
6.7	Operational monitoring	113	9.4.2	Valve gear - Lubrication	200
6.8	Emission label - Check	114	9.4.3	Valve clearance - Check and adjustment	201
6.9	Engine - Stopping in manual mode (test run)	115	9.4.4	Cylinder head cover - Removal and installation	204
6.10	Engine - Emergency stop	117	9.5	Injection Pump / HP Pump	205
6.11	After stopping the engine - Engine remains ready for operation	118	9.5.1	HP pump - Filling with engine oil	205
6.12	After shutting down engine - Putting engine out of operation	119	9.6	Injector	208
6.13	Plant - Cleaning	120	9.6.1	Injector - Replacement	208
			9.6.2	Injector - Removal and installation	209
7	Troubleshooting		9.7	Fuel System	215
7.1	Troubleshooting	122	9.7.1	Fuel system - Venting	215
7.2	Engine Control Unit ADEC (ECU 7) for Series 4000 genset engines - Fault messages	125	9.8	Fuel Filter	217
8	Maintenance		9.8.1	Fuel filter - Replacement	217
8.1	Maintenance task reference table [QL1] for MS50176/04, MS50232/03, MS50233/03	174	9.8.2	Fuel prefilter - Pressure ratios	218
8.2	Maintenance task reference table [QL1] for MS50177/04, MS50178/03, MS50179/04, MS50188/04, MS50189/03, MS50190/04, MS50191/04, MS50230/03, MS50231/03	176	9.8.3	Fuel prefilter (non-switchable) - Draining	219
8.3	Maintenance task reference table [QL1] for MSE50188/04, MSE50190/04, MSE50230/03, EPA regions	178	9.8.4	Fuel prefilter (switchable) - Draining	222
8.4	Maintenance task reference table [QL1] for MSE50189/02, EPA regions	180	9.8.5	Fuel prefilter (switchable) - Low pressure indicator check	225
8.5	Maintenance task reference table [QL1] for MSE50270/02, EPA regions	181	9.8.6	Fuel prefilter (non-switchable) - Filter element replacement	226
9	Task Description		9.8.7	Fuel prefilter (switchable) - Filter element replacement	228
9.1	Engine	183	9.8.8	Fuel prefilter - Venting	231
9.1.1	Barring engine manually	183	9.9	Charge-Air Cooling	233
9.1.2	Engine - Turning on starting system	185	9.9.1	Intercooler - Checking condensate drain for coolant leak and obstruction	233
9.1.3	Engine - Test run	186	9.10	Air Filter	234
9.2	Cylinder Liner	187	9.10.1	Air filter - Replacement	234
9.2.1	Cylinder liner - Endoscopic inspection	187	9.10.2	Air filter - Check	235
9.2.2	Cylinder liner - Instructions and comments on endoscopic and visual examination	189	9.10.3	Air filter - Removal and installation	236
9.3	Crankcase Breather	191	9.11	Air Intake	237
9.3.1	Crankcase breather - Oil mist separator replacement	191	9.11.1	Contamination indicator - Signal ring position check	237
9.3.2	Crankcase breather - Oil separator element replacement	192	9.12	Starting Equipment	238
9.3.3	Crankcase breather - Filter element replacement	195	9.12.1	Air starter - Manual operation	238
9.4	Valve Drive	196	9.13	Lube Oil System, Lube Oil Circuit	239
			9.13.1	Engine oil - Change	239
			9.13.2	Engine oil - Level check	241
			9.13.3	Engine oil - Sample extraction and analysis	242
			9.13.4	Oil priming pump - Visual inspection	244
			9.14	Oil Filtration / Cooling	245
			9.14.1	Engine oil filter - Replacement	245
			9.14.2	Oil indicator filter - Cleaning and check	246
			9.14.3	Centrifugal oil filter - Cleaning and filter sleeve replacement	249
			9.15	Coolant Circuit, General, High-Temperature Circuit	252
			9.15.1	Engine coolant - Level check	252
			9.15.2	Engine coolant - Change	254
			9.15.3	Engine coolant - Draining	255

9.15.4	Engine coolant - Filling	257	9.19	Engine Mounting / Support	291
9.15.5	Engine coolant pump - Relief bore check	260	9.19.1	Engine mounting - Checking resilient element	291
9.15.6	Engine coolant - Sample extraction and analysis	261	9.19.2	Engine mounting - Checking securing screws for firm seating	292
9.15.7	Fuel cooler - External contamination and leak-tightness check	263	9.19.3	Resilient mounts of generator mounting - Check	294
9.15.8	Fuel cooler - Cleaning	264	9.19.4	Generator mounting - Securing screw firm seating check	295
9.15.9	Preheater - Overview	266	9.20	Wiring (General) for Engine/Gearbox/Unit	296
9.15.10	Preheater - Function and leak check	267	9.20.1	Engine cabling - Check	296
9.15.11	Preheater - Overhaul	268	9.21	Accessories for (Electronic) Engine Governor / Control System	297
9.15.12	Preheater - Solenoid switch replacement	269	9.21.1	Engine governor and connectors - Cleaning	297
9.15.13	Preheater - Heating element and housing check	271	9.21.2	Engine governor plug connections - Check	298
9.15.14	Preheater thermostats - Replacement	274	9.21.3	Engine governor ECU 7 - Removal and installation	299
9.15.15	Coolant expansion tank - Valve cover replacement	277	10	Appendix A	
9.16	Low-Temperature Circuit	279	10.1	Abbreviations	300
9.16.1	Charge-air coolant level - Check	279	10.2	Contact person/Service partner	304
9.16.2	Charge-air coolant - Change	281	11	Appendix B	
9.16.3	Charge-air coolant - Draining	282	11.1	Standard Tools and Special Tools	305
9.16.4	Charge-air coolant - Filling	284	11.2	Index	314
9.16.5	Charge-air coolant pump - Relief bore check	287			
9.17	Belt Drive	288			
9.17.1	Drive belt - Condition check	288			
9.18	Battery-Charging Generator	289			
9.18.1	Battery-charging generator - Drive belt tension adjustment	289			
9.18.2	Battery-charging generator drive - Drive belt replacement	290			

1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by Rolls-Royce Solutions come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations

Prior to operation, make sure that the latest version is used. The latest version can be found at: <http://www.mtu-solutions.com>

2.2 Intended use of gendrive engines

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the admissible operating parameters in accordance with the (→ technical sales documents – product data – TVU data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/Rolls-Royce Solutions contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques

The product shall not be operated in explosive atmospheres unless it fulfills the conditions for such use and approval has been granted.

The section below applies to applications deployed in regions under the jurisdiction of the EPA.

- Use of EPA Tier 2 and Tier 3 certified genset engines.
- EPA Tier 2 and Tier 3 certified genset engines may only be used in generator sets for stationary emergency applications. Non-emergency operation is limited to 100 hrs per year.

The engines may be used in fixed speed engine-generator sets only. Permanent limitation to a max. engine power of less than 560 kW (e.g. by means of correspondingly low generator power) is not admissible.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.